

Purpose

We want a framework that allows us to model the forward rate $f(t, T)$ directly.

$f(t, T)$ - The rate that can be locked in at time t for a loan that is infinitesimally short.

We assume very general dynamics: $f(t, T) = -\partial_T \log(P(t, T))$

$$f(t, T) = f(0, T) + \int_0^t \alpha(s, T) ds + \int_0^t \sigma(s, T) dW(s)$$

With the assumption that α and σ are progressive and non-explosive.

Why do this? We can now have different dynamics for bonds with different maturities.

Risk neutral dynamics of forward rate is fully specified by σ :

$$f(t, T) = f(0, T) + \int_0^t \left(\sigma(s, T) \int_s^T \sigma(s, u)^T du \right) ds + \int_0^t \sigma(s, T) dW^*(s)$$

Issues with HJM

HJM is a generalized short rate model: Possibly too general.

Infinite-dimensional: $\nexists$ a deterministic function g and Markov process $X(t)$ such that:

$$f(t, T) = g(t, T, X(t)), \forall T > t$$

One fix: $f(t, T)$ has a finite-dimensional affine representation.

Second fix: We can enforce certain conditions on σ , that correspond to enforcing markovianity.

Why do we want Markovianity?

$$dr(t) = \zeta^*(t)dt + \sigma(t, t)dW^*(t)$$

$$\begin{aligned} \zeta^*(t) = & \partial_t f(0, t) + \int_0^t \partial_t \sigma(s, t) \int_s^t \sigma(s, u)^\top du ds \\ & + \int_0^t \sigma(s, t) \sigma(s, t)^\top ds + \int_0^t \partial_t \sigma(s, t) dW^*(s) \end{aligned}$$

Enforcing Markovianity

We assume the following volatility dynamics:

$$\sigma(s, t) = (\sigma_1(s, t), \dots, \sigma_d(s, t)) \quad \kappa = \text{deterministic function}$$

$$\sigma_i(s, t) = \eta_i(s) e^{-\int_s^t \kappa_i(u) du}, i = 1, \dots, d \quad \eta = \text{progressive process}$$

Now define:

$$y_i(t) = \int_0^t \sigma_i^2(s, t) ds \quad dy_i(t) = (\eta_i^2(t) - 2\kappa_i(t)y_i)dt, \quad y_i(0) = 0$$

$$x_i(t) = \int_0^t \sigma_i^*(s, t) ds + \int_0^t \sigma_i(s, t) dW_i^*(s) \quad dx_i(t) = (y_i(t) - \kappa_i(t)x_i(t))dt + \eta_i(t)dW^*(t), \quad x_i(0) = 0$$

$$r(t) = f(0, t) + \sum_{i=1}^d x_i(t)$$

$$f(t, T) = f(0, t) + \sum e^{-\int_t^T \kappa_i(s) ds} (x_i(t) + \gamma_i(t, T)y_i(t)) \quad \gamma_i(t, T) := \int_t^T \exp(-\int_t^u \kappa_i(s) ds) du$$

$$P(t, T) = \frac{P(0, T)}{P(0, t)} \exp \left(-\frac{1}{2} \sum_{i=1}^d \gamma_i(t, T)^2 y_i(t) - \sum_{i=1}^d \gamma_i(t, T) x_i(t) \right)$$

Much easier to simulate!

Affine Models are just a specific case of Markovian HJM

Consider an Affine model with state dynamics:

$$dX(t) = \mu(t, X(t))dt + \rho(t, X(t))dW^*(t)$$

Remember Affine assumption:

$$P(t, T) = \mathbb{E}_t^{\mathbb{Q}}[e^{-\int_t^T R(X_s)ds}] = \exp(\phi(t, T) + \psi(t, T)X_t)$$

Which implies:

$$\begin{aligned} f(t, T) &= -\partial_T \phi(t, T) - \partial_T \psi(t, T)X_t \\ df(t, T) &= (\dots)dt - \partial_T \psi(t, T)dX(t) \\ &= (\dots)dt - \partial_T \psi(t, T)\rho(t, X_t)dW^*(t) \end{aligned} \quad \sigma(t, T) = -\partial_T \psi(t, T)\rho(t, X_t)$$

So really, from HJM perspective, affine model is just Markovian HJM with:

$$\eta_i(t) = \{\rho(t, X_t)\}_i, \quad e^{-\int_t^T \kappa_i(u)du} = \{-\partial_T \psi(t, T)\}_i, \quad i = 1, \dots, d$$

Option pricing - Andreasen (2001)

$$V = V(t, x(t), y(t)) = \mathbb{E}_t^Q[e^{-\int_t^T r(s)ds} \Phi(x(T), y(T))]$$

Not a Martingale. But discounted version is.

$$0 = \partial_t V - [f(0, t) + x]V + [-\kappa(t)x + y]\partial_x V + \frac{1}{2}\eta(t, x, y)^2 \partial_x^2 V \\ + (\eta(t, x, y)^2 + 2\kappa(t)y) \partial_y V, \quad V(T, x, y) = \Phi(x, y)$$

Let's see it for a caplet with constant κ and $\eta(t, x, y) := \sigma F(t, T, T + \delta)^\alpha$

$$Cpl(0) = \delta \mathbb{E}_0^Q[e^{-\int_0^{T+\delta} r(s)ds} ((F(T, T + \delta) - K)^+)]$$

$$Cpl(0) = \mathbb{E}_0^Q[e^{-\int_0^T r(s)ds} (1 - P(T, T + \delta)\tilde{K})^+], \quad \tilde{K} = 1 + \delta K$$

We can then just find the price using Feynman-Kac and the contract function:

$$\Phi(t, x(t), y(t)) = (1 - P(T, T + \delta)\tilde{K})^+ \tag{76}$$

$$= \left[1 - \tilde{K} \frac{P(0, T + \delta)}{P(0, T)} \exp \left(-\frac{1}{2} \gamma(T, T + \delta)^2 y(t) - \gamma(T, T + \delta) x(t) \right) \right]^+ \tag{77}$$